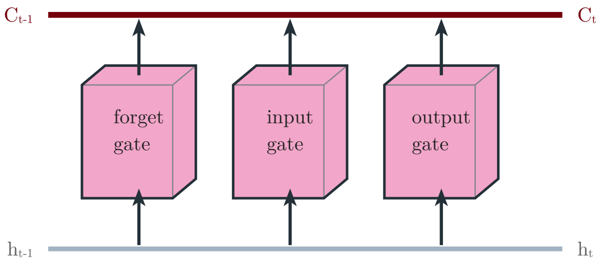
LSTM memory cell

cell state C_t (top conveyor) · hidden state h_t (bottom) · three gates



forget $f_t = \sigma\big(W_f[h_{t-1}, x_t] + b_f\big)$

input $i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$

candidate $\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$

output $o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \qquad h_t = o_t \odot \tanh(C_t)$$